Q1 **Palindrome number/string** 10 marks

Palindrome number/string

Your Code

```
import java.util.Scanner;
public class PalindromeString
{
    public static void main(String[]args)
    {
        String str="madam";
        String rev="";
        for(int i=str.length()-1;i>=0;i--)
        {
            rev=rev+str.charAt(i);
        }
        if(str.equals(rev))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

Input

stdin input

Output

```
Palindrome
```

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Q2 **First n perfect numbers** 10 marks

First n perfect numbers

Your Code

```
import java.util.Scanner;
public class PerfectNumber
{
    public static void main(String[]args)
    {
        int a[]={6,10,28,496,20};
        System.out.println("Perfect Numbers in the array:\n");
        for(int j=0;j<a.length;j++)
        {
            int sum=0;
            for(int i=1;i<=a[j]/2;i++)
            {
                if(a[j]%i==0)
                    sum=sum+i;
            }
            if(sum==a[j])
            {
            System.out.println(a[j]);
            }
        }
    }
}
```

Input

stdin input

Output

```
Perfect Numbers in the array:

6
28
496
```

Q3 **Decimal number equivalent to binary number and octal number** 10 marks

Decimal number equivalent to binary number and octal number

Your Code

```java
import java.util.Scanner;
public class DecimalArray
{
    public static void main(String[] args)
    {
        int a[] = {10, 15, 20, 25};
        for (int i = 0; i < a.length; i++)
        {
            System.out.println("Decimal = " + a[i]);
            System.out.println("Binary = " + Integer.toBinaryString(a[i
            System.out.println("Octal = " + Integer.toOctalString(a[i])
            System.out.println();
        }
    }
}
```

Input

```
stdin input
```

Output

```
Decimal = 10
Binary = 1010
Octal = 12

Decimal = 15
Binary = 1111
```

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Q4 **Nth max number and nth min number in array then sum of it and difference of it** 10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```java
import java.util.Arrays;
public class NthMaxMin
{
    public static void main(String[] args)
    {
        int a[] = {10, 50, 30, 20, 40};
        int n = 2;
        Arrays.sort(a);
        int nthMin = a[n - 1];
        int nthMax = a[a.length - n];
        System.out.println("Nth Minimum = " + nthMin);
        System.out.println("Nth Maximum = " + nthMax);
        System.out.println("Sum = " + (nthMin + nthMax));
        System.out.println("Difference = " + (nthMax - nthMin));
    }
}
```

Input

```
stdin input
```

Output

```
Nth Minimum = 20
Nth Maximum = 40
Sum = 60
Difference = 20
```

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Q5 Find the average positive and negative numbers upto -1

Find the average positive and negative numbers upto -1

Your Code

```java
import java.util.Scanner;
public class AverageNumbers
{
    public static void main(String[] args)
    {
        int[] a = {10, 20, -5, 15, -8, -1};
        int posSum = 0, negSum = 0;
        int posCount = 0, negCount = 0;
        for (int i = 0; i < a.length; i++)
        {
            if (a[i] == -1)
                break;
            if (a[i] >= 0)
            {
                posSum += a[i];
                posCount++;
            }
            else
            {
                negSum += a[i];
                negCount++;
            }
        }
        System.out.println("Average of Positive Numbers = " + (posSum
        System.out.println("Average of Negative Numbers = " + (negSum
    }
}
```

Input

stdin input

Output

```
Average of Positive Numbers = 15
Average of Negative Numbers = -6
```

Q6 Removing duplicates in array

Removing duplicates in array

```java
import java.util.Scanner;
public class RemoveDuplicates
{
    public static void main(String[] args)
    {
        int a[] = {1, 2, 3, 2, 4, 1, 5};
        System.out.println("Array after removing duplicates:");
        for (int i = 0; i < a.length; i++)
        {
            boolean duplicate = false;
            for (int j = 0; j < i; j++)
            {
                if (a[i] == a[j])
                {
                    duplicate = true;
                    break;
                }
            }
            if (!duplicate)
            {
                System.out.print(a[i] + " ");
            }
        }
    }
}
```

stdin input

Output

```
Array after removing duplicates:
1 2 3 4 5
```

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Q7 Matrix multiplication

Matrix multiplication

```java
public class MatrixMultiplication
{
    public static void main(String[] args)
    {

        int a[][] = {{1, 2}, {3, 4}};
        int b[][] = {{5, 6}, {7, 8}};
        int c[][] = new int[2][2];
        for (int i = 0; i < 2; i++)
        {
            for (int j = 0; j < 2; j++)
            {
                c[i][j] = 0;
                for (int k = 0; k < 2; k++)
                {
                    c[i][j] = c[i][j] + a[i][k] * b[k][j];
                }
            }
        }
        System.out.println("Result Matrix:");
        for (int i = 0; i < 2; i++)
        {
            for (int j = 0; j < 2; j++)
            {
                System.out.print(c[i][j] + " ");
            }
            System.out.println();
        }
    }
```

Output

```
Result Matrix:
19 22
43 50
```